

Assignment- Week 0

Part E — Reflection

1. Which concept from Week 0 you found hardest, and what specifically made it click — or what still has not clicked.

The concept from Week 0 that I found hardest was **understanding how Flexbox and Grid work for webpage layout**. Since web development is still quite new to me, I initially found it confusing to understand how different elements were positioned and how the layout could change for different screen sizes. It started to make more sense when I actually used Flexbox for my navigation and Grid for the main page layout and project cards. Testing my page at 360px, 768px and 1440px helped me see how the layout responds to different screen sizes. What made it click for me was being able to change the CSS and immediately see the result on the webpage. I am still learning these concepts, especially because I have recently started learning React, but this assignment made them much more interesting and easier to understand through practice.

2. One bug you hit during this assignment: the symptom, how you diagnosed it, and the actual cause. Use the debugging procedure in Chapter 12.5 and show that you followed it rather than guessing. A bug you fixed by trial and error is fine to report, as long as you say so.

One bug I faced during this assignment was while testing my page at smaller screen sizes. At first, some parts of the page looked like they were not fitting properly, especially when I was checking the 360px responsive view. I first reproduced the issue by setting the viewport to 360px and checking the page again. Following the debugging procedure from Chapter 12.5, I then inspected the layout in DevTools and checked the actual viewport size and browser settings instead of immediately changing my CSS. I noticed that the page was being viewed at a different zoom level, which made some parts appear more cropped than they actually were. I changed the zoom to 100% and tested the page again. The layout then displayed correctly. After that, I checked the page again at 360px, 768px and 1440px to make sure the problem was actually fixed. This experience taught me to check my assumptions and test one change at a time instead of changing different CSS rules randomly.

3. One design decision where you had a genuine choice — Flexbox or Grid, a class toggle or an inline style, delegation or individual listeners — and your reasoning.

One design decision I made was choosing **Grid for the main page layout instead of using Flexbox for everything**. I used Flexbox for my navigation because it was useful for keeping the website name on one side and the navigation links on the other side. For the main page, I chose Grid because I had different sections such as Home, About, Technical Knowledge, Programme, Projects and Enquiry, and Grid made it easier to organise these areas into a clear structure. I also

used Grid for the project cards so that the number of cards in a row can adjust automatically depending on the screen size. I chose this approach because it made the page easier to manage and also helped me create a responsive layout for mobile, tablet and desktop screens.

4. Your completed Exercise 13.1 checklist.

I completed the main requirements of Exercise 13.1 through my personal portfolio and enquiry page. I created a single-page website with header navigation, an About section, technical skills, a projects area and an enquiry form. I used semantic HTML elements such as header, nav, main, section, aside, article and footer to give the page a clear structure. I used **Flexbox for the navigation** and **CSS Grid for the main layout and project cards**, including the responsive auto-fit Grid layout.

The page is **mobile-first and responsive**, and I tested it at **360px, 768px and 1440px**. I also added fragment navigation so the menu links move to the correct sections. The enquiry form includes native validation as well as custom JavaScript validation with inline error messages, and the page does not reload when the form is submitted. I added a **dark mode toggle using a class on the body**, with the preference stored in localStorage.

For JavaScript, I used FormData and Object.fromEntries(), event delegation with closest(), an array of project data, createElement() and textContent. I also tested keyboard behaviour, including the Escape key for clearing the form. I checked the browser console during normal use and did not find JavaScript errors.

Overall, I completed the Exercise 13.1 requirements and tested the main features rather than only checking that the page looked correct.

5. Your Lighthouse accessibility score and validator result.

After completing my webpage, I tested it using Chrome Lighthouse and the W3C HTML Validator. My **Lighthouse Accessibility score was 100/100**, which was a good result because accessibility was one of the main areas I was focusing on while building the page. The overall Lighthouse results were **93 for Performance, 100 for Accessibility, 100 for Best Practices and 91 for SEO**.

I also checked my index.html using the **W3C Nu HTML Checker**. The validator showed **0 errors and 0 warnings**, which confirmed that my HTML structure was valid. I was happy with these results because they showed that the page was not only working visually, but was also properly structured and accessible.